The brain clock

And its neurological implications

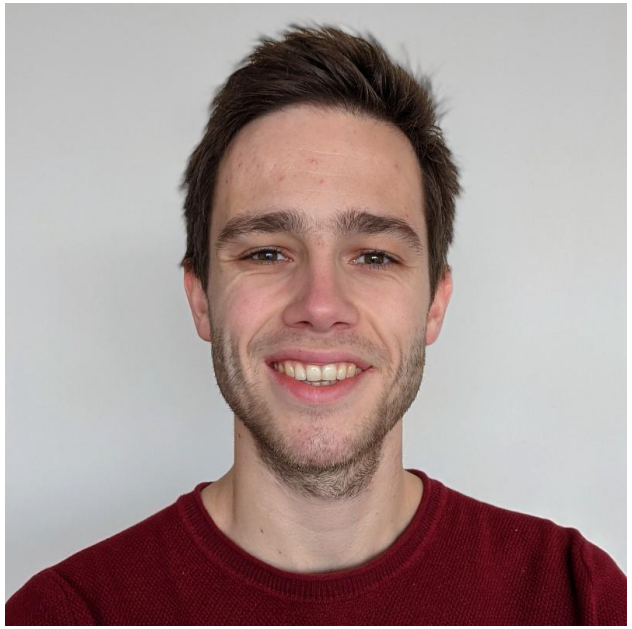
Disclosures

This work has partly been performed during the industrial PhD of Stijn Denissen, funded by a Baekeland grant of Flanders Innovation and Entrepreneurship. The PhD was a collaboration between the AIMS lab (VUB) and ico**metrix**.

The federated learning work was made possible by a IOF-POC grant from the VUB and a travel grant from the FWO Flanders.

Stijn is currently a junior postdoctoral fellow of the FWO Flanders.

Predicting age from images

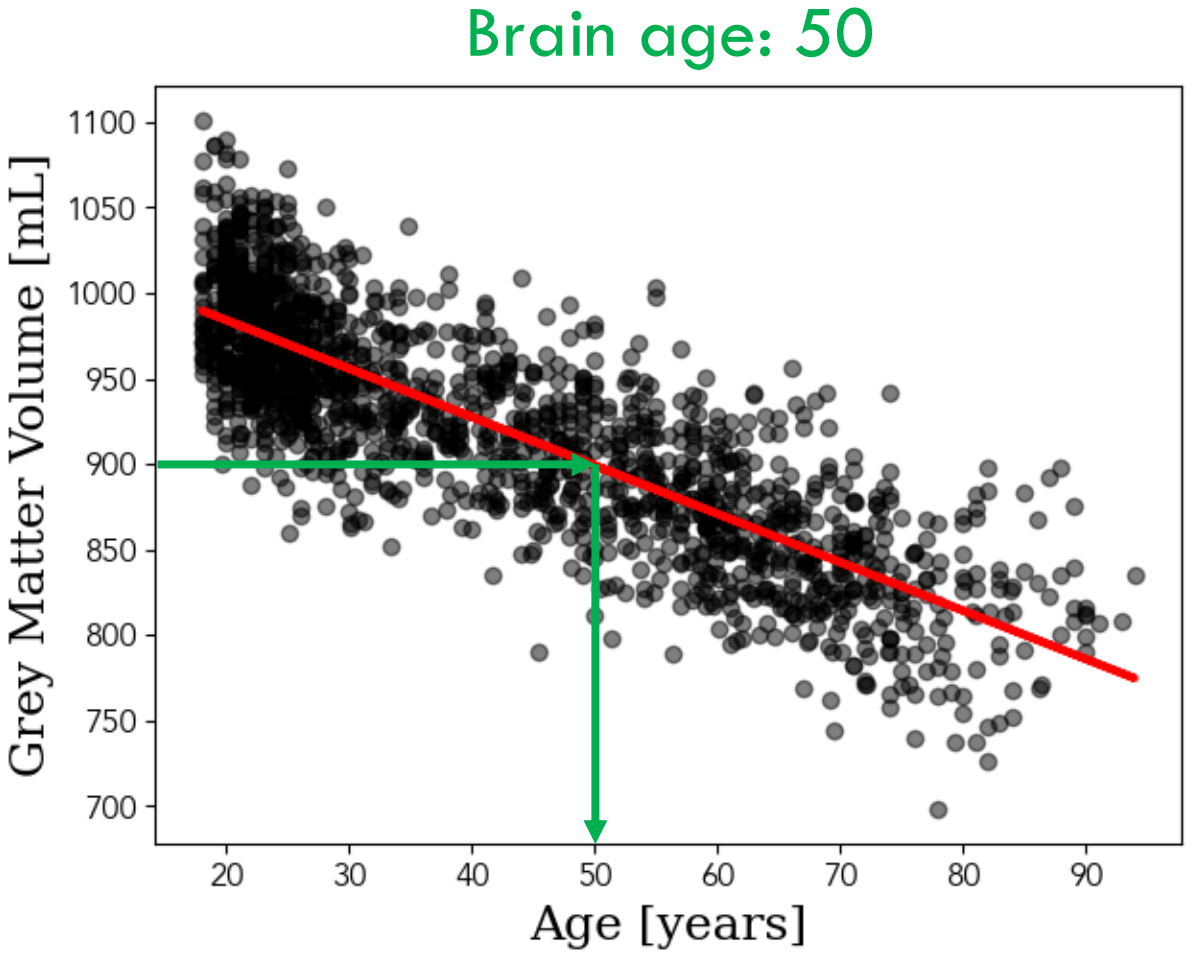


wrinkles ↑
receding hairline

grey hair ↑
….

Images can mislead you

So can brain images



Brain Age Gap (BAG)

= brain age – chronological age

BAG = 0	Healthy Controls
BAG –	Brain looks younger
BAG +	Brain looks older

→ E.g. neurological disorders

Image created from publicly available HC data

Part 1
The technical tale

The brain clock

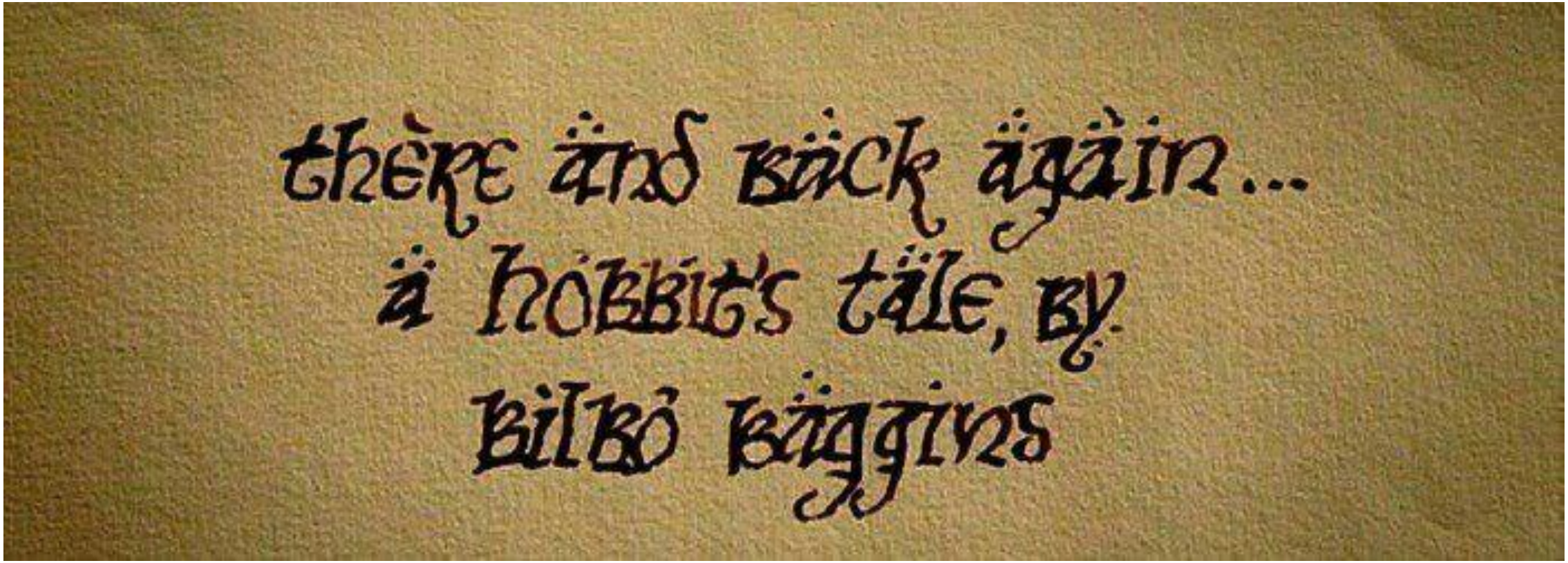
And its neurological implications

Part 2
The clinical tale

Part 1

The technical tale

Simple to complicated (and back to simple?)



Level 1: 1D linear model

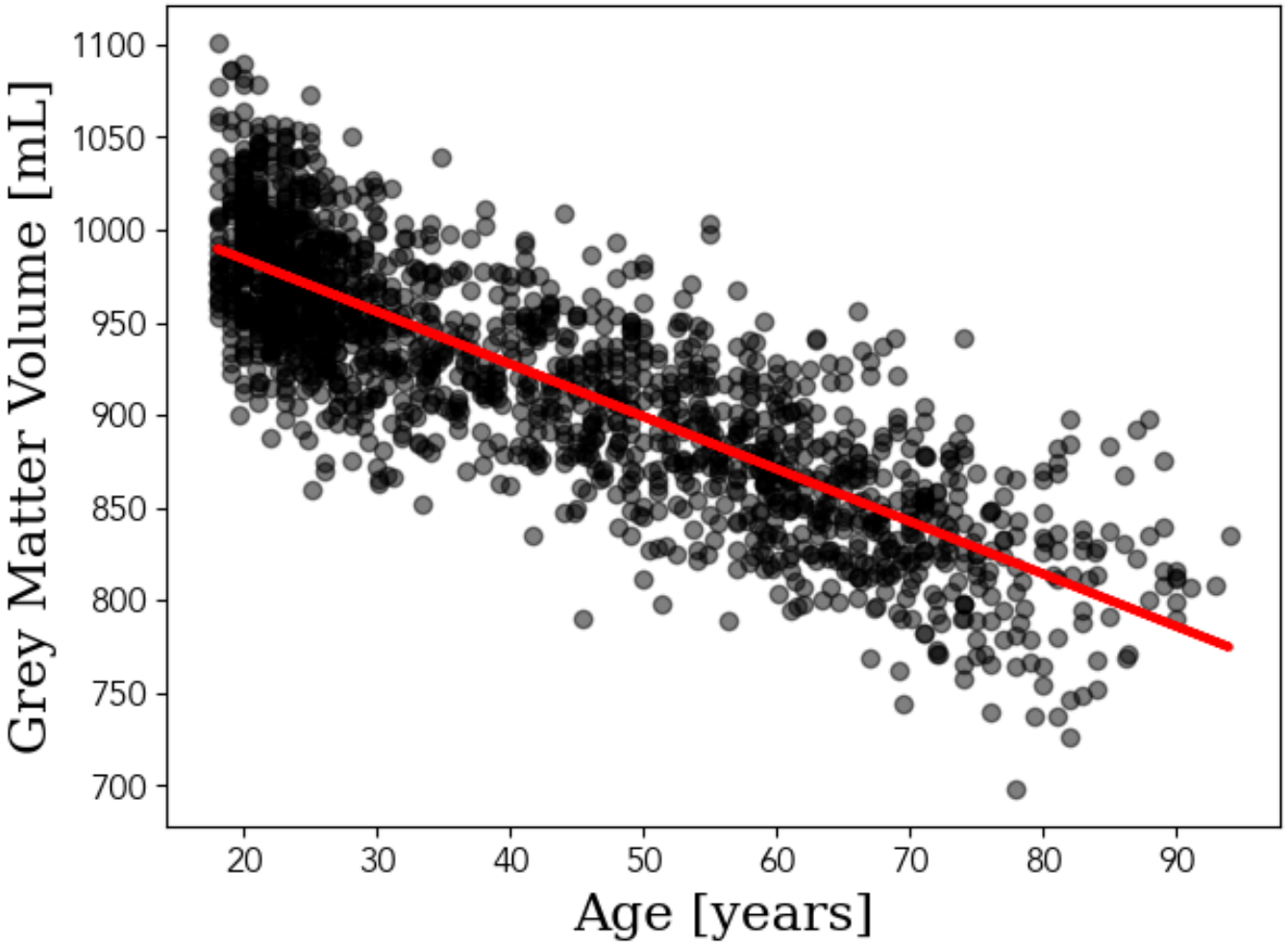
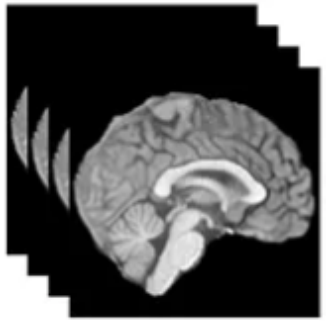


Image created from publicly available HC data

Level 2: 12-D linear model



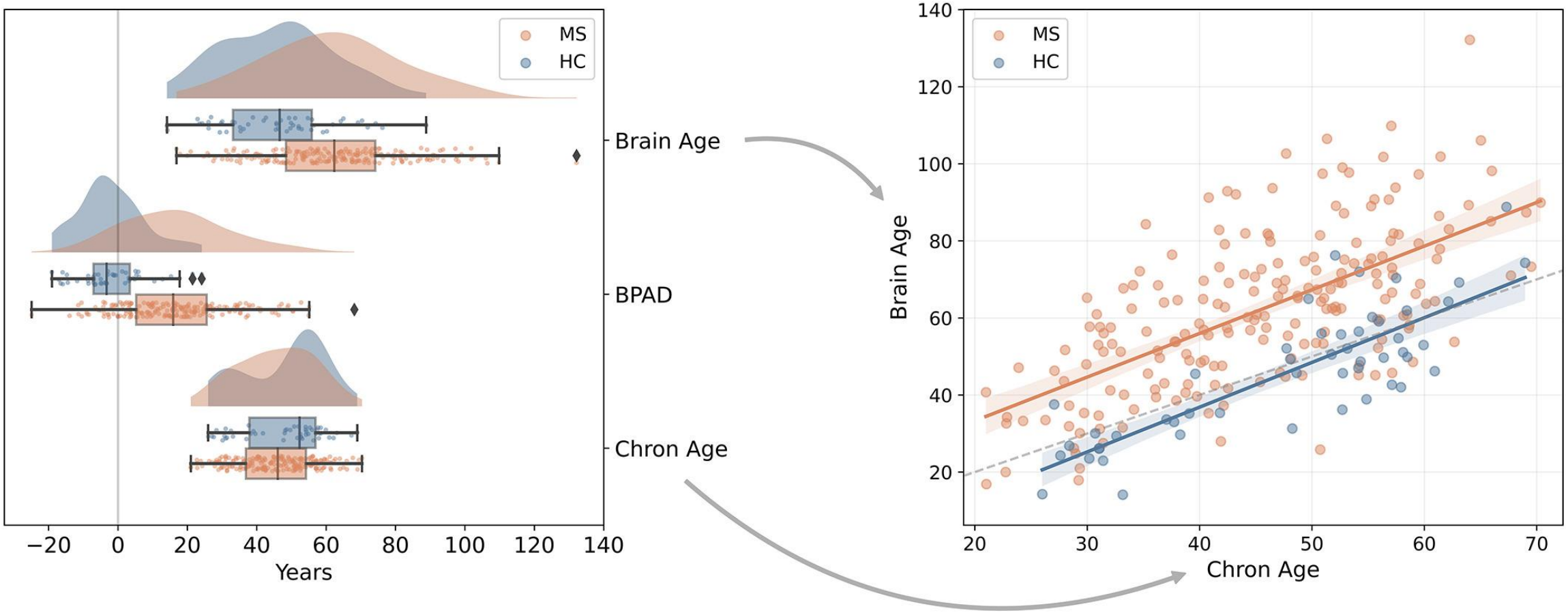
Segment →

Feature	Weight	Standard Error	*t*	*p*
Intercept	41.8867	0.216	193.509	<0.001
Grey matter	−7.5859	1.021	−7.427	<0.001
White matter	−0.8746	0.269	−3.252	0.001
Lateral ventricles	2.2432	0.359	6.244	<0.001
Cortical grey matter—frontal lobe	−4.0462	0.628	−6.438	<0.001
Cortical grey matter—occipital lobe	0.4299	0.314	1.370	0.171
Cortical grey matter—temporal lobe	−1.0626	0.403	−2.638	0.008
Cortical grey matter—parietal lobe	−1.0037	0.424	−2.366	0.018
Hippocampus—left	0.2280	0.304	0.750	0.453
Hippocampus—right	1.4228	0.311	4.573	<0.001
Thalamus—left	−2.1227	0.526	−4.036	<0.001
Thalamus—right	−2.5859	0.511	−5.057	<0.001
Sex	−3.4668	0.229	−15.136	<0.001

Denissen et al. 2022, Eur. J. Neurol.

Level 2: 12-D linear model

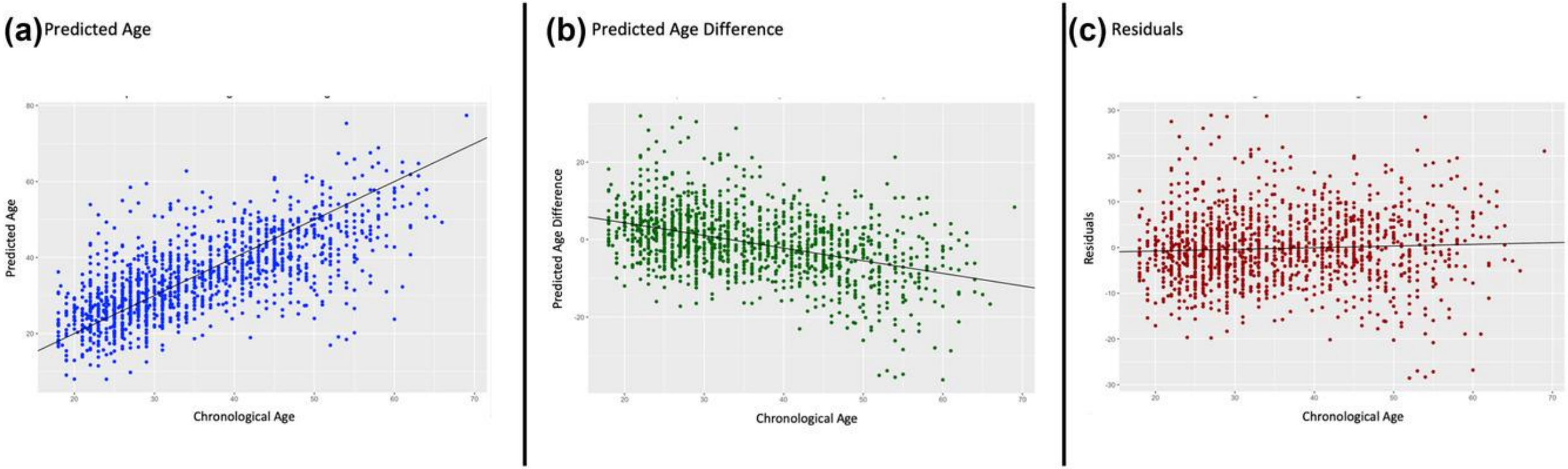
MAE_{HC}: 7.85 year



Denissen et al. 2022, Eur. J. Neurol.

Level 3: N-D Gaussian Processes Regression

MAE_{HC}: 5.68 year

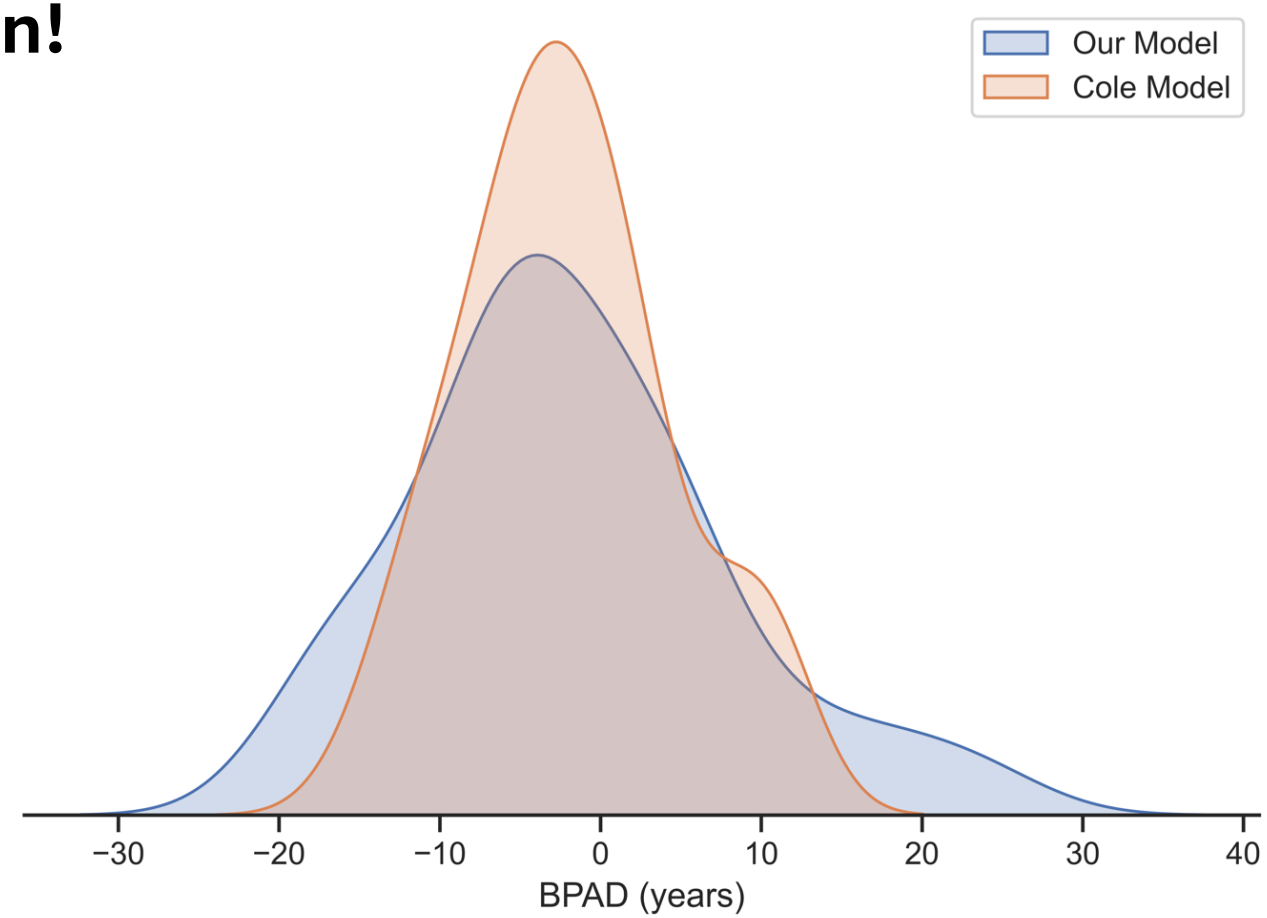


Clausen et al. 2021. Brain and Behavior

Level 3: N-D Gaussian Processes Regression

❤️ **Good generalisation!**

MAE_{HC}: 5.52 years

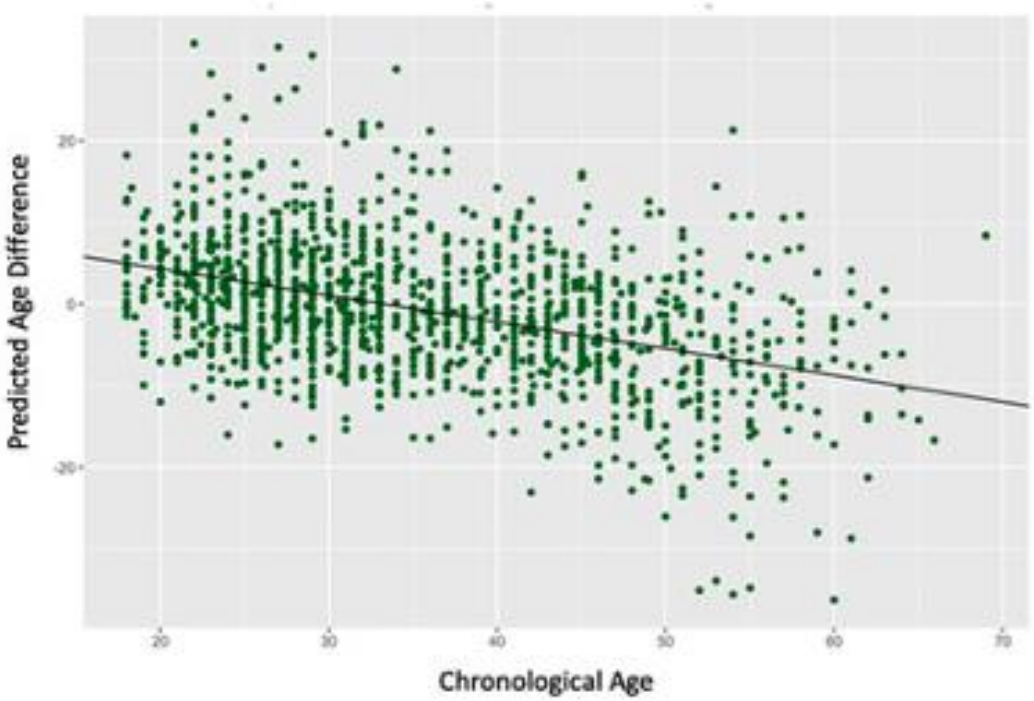


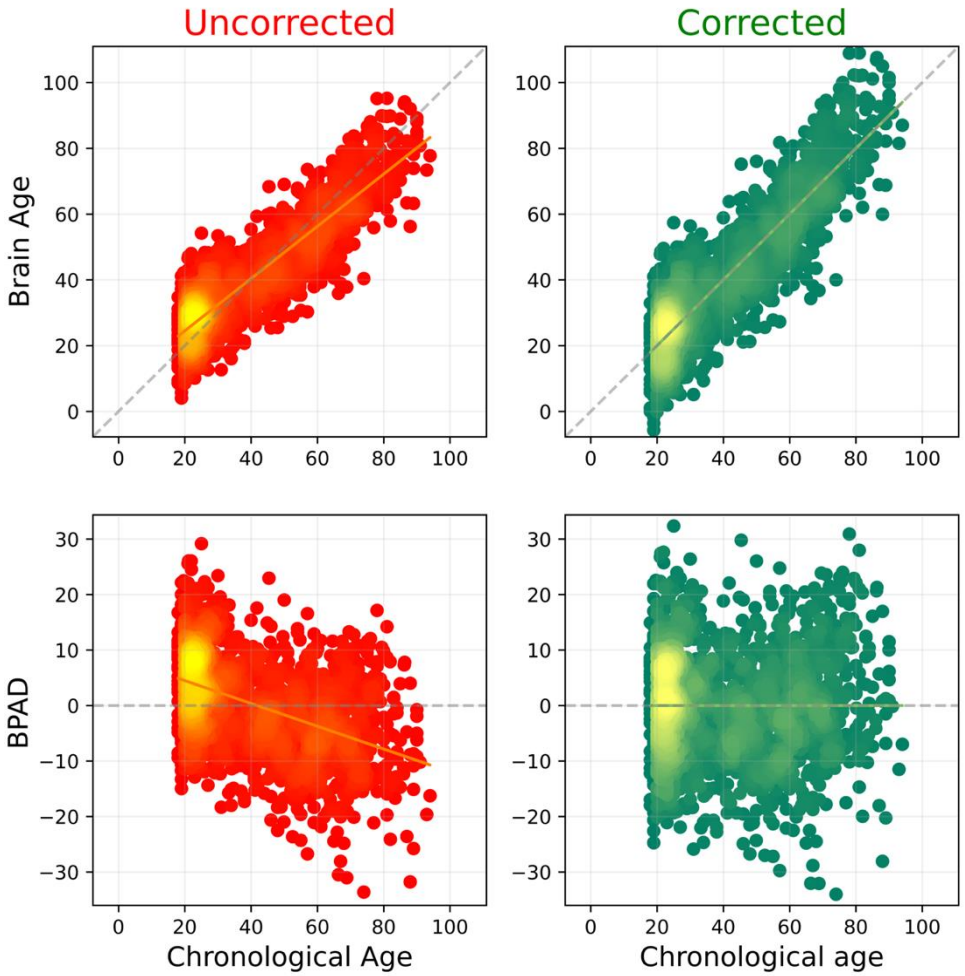
(= Clausen 2021 model)

Denissen et al. 2022, Eur. J. Neurol.

Level 3: N-D Gaussian Processes Regression

⚠ **Inherent age bias!**

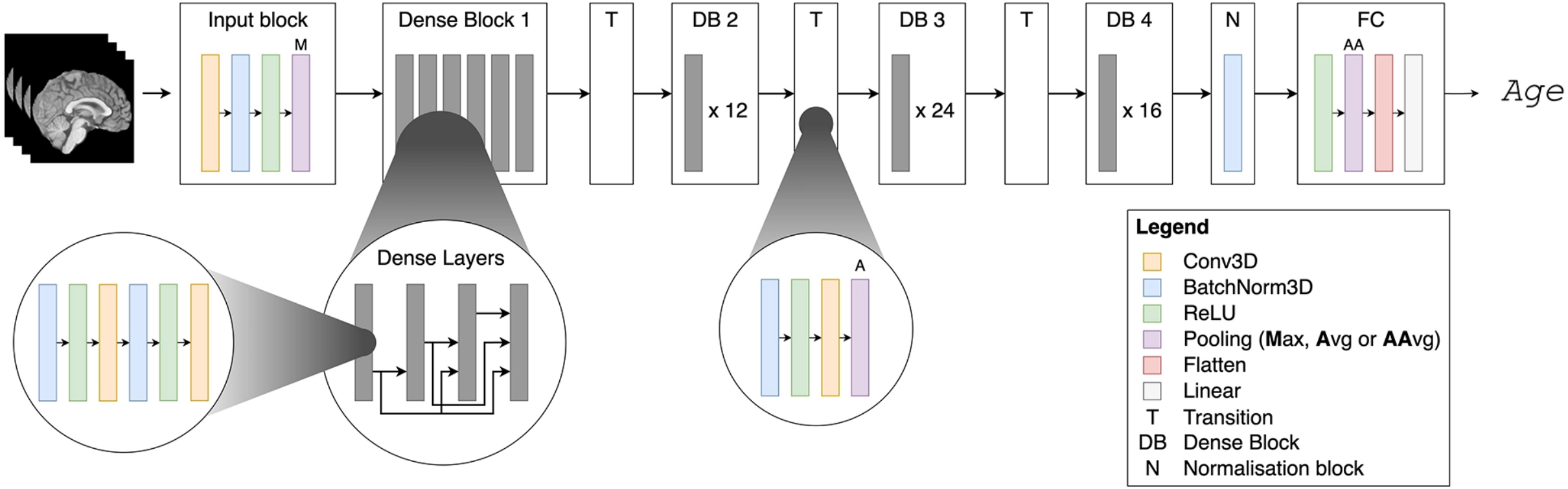




Clausen et al. 2021. Brain and Behavior

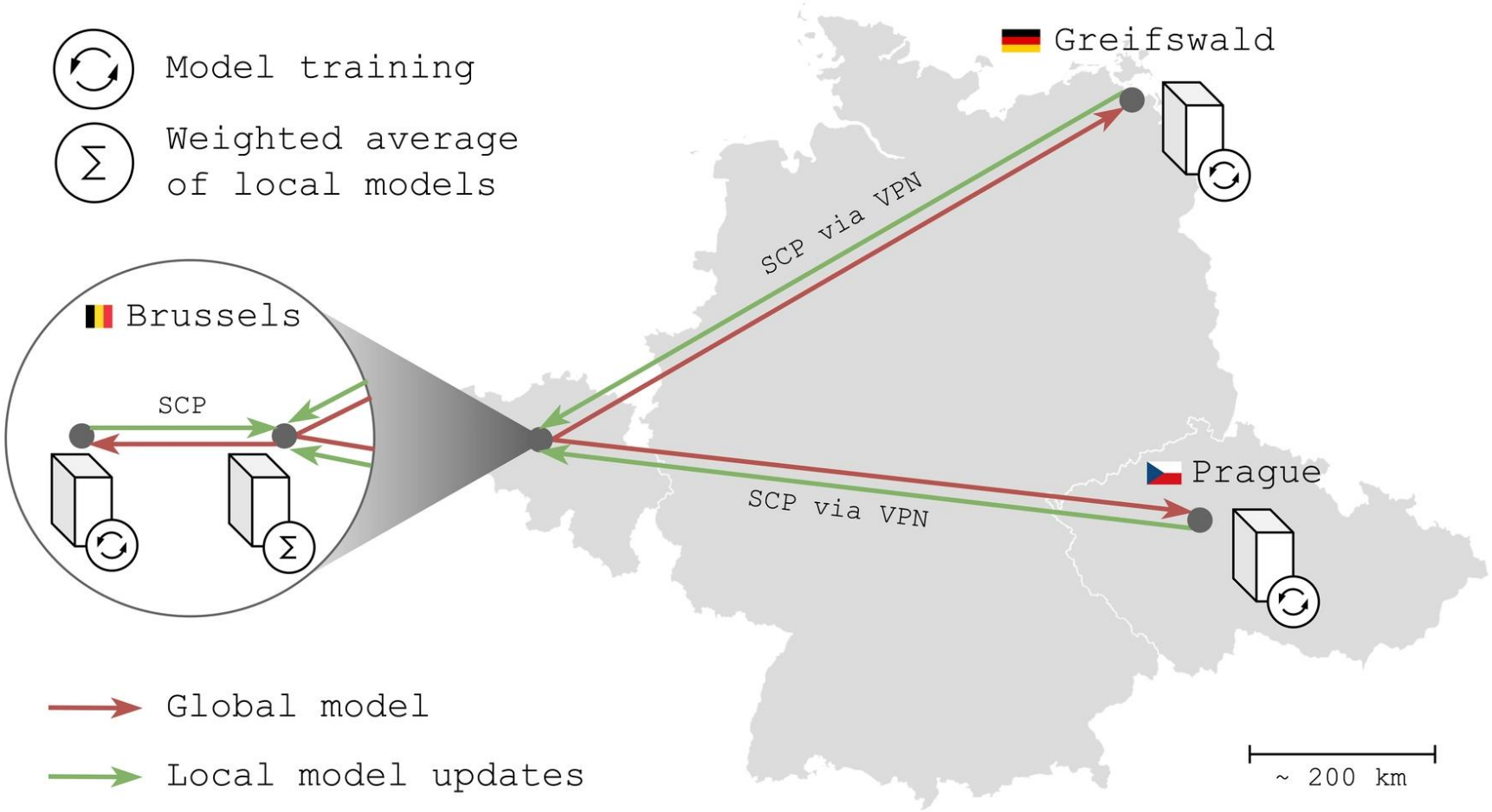
Denissen et al. 2022, Eur. J. Neurol.

Level 4: Deep CNNs



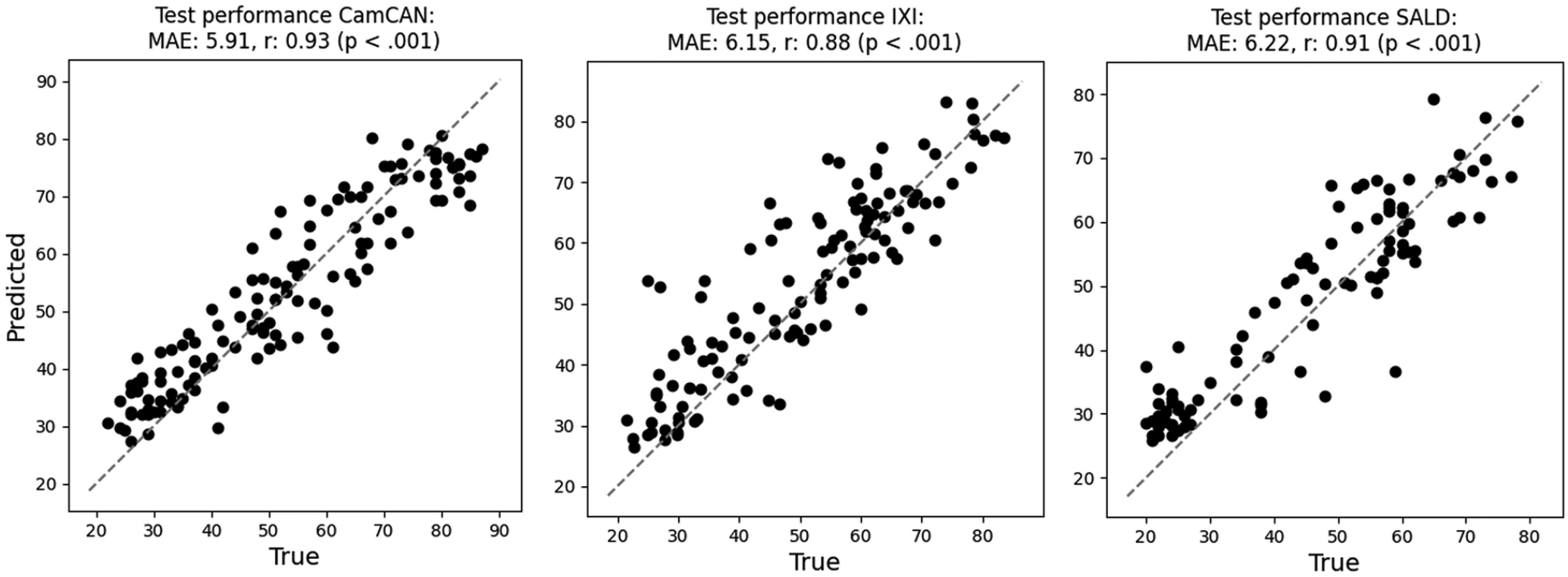
Denissen et al. 2026, Front. Digit. Health

Level 4: Deep CNNs



Denissen et al. 2026, Front. Digit. Health

Level 4: Deep CNNs

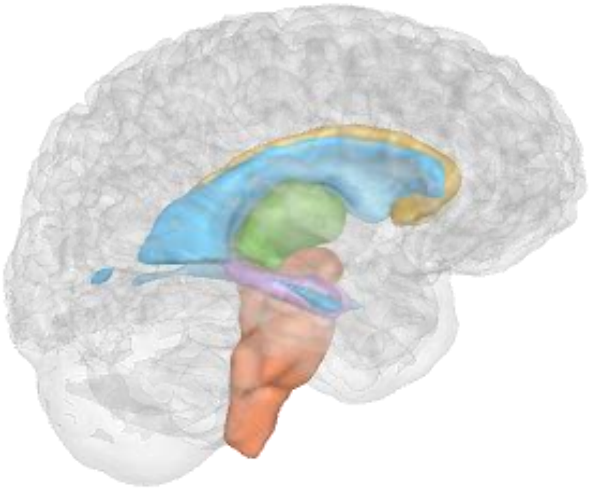


This was only a federated learning proof-of-concept.
CNNs can do much better!

Denissen et al. 2026, Front. Digit. Health

Level 4: Deep CNNs

- How old is my brain according to this model?
- https://stijndenissen.me



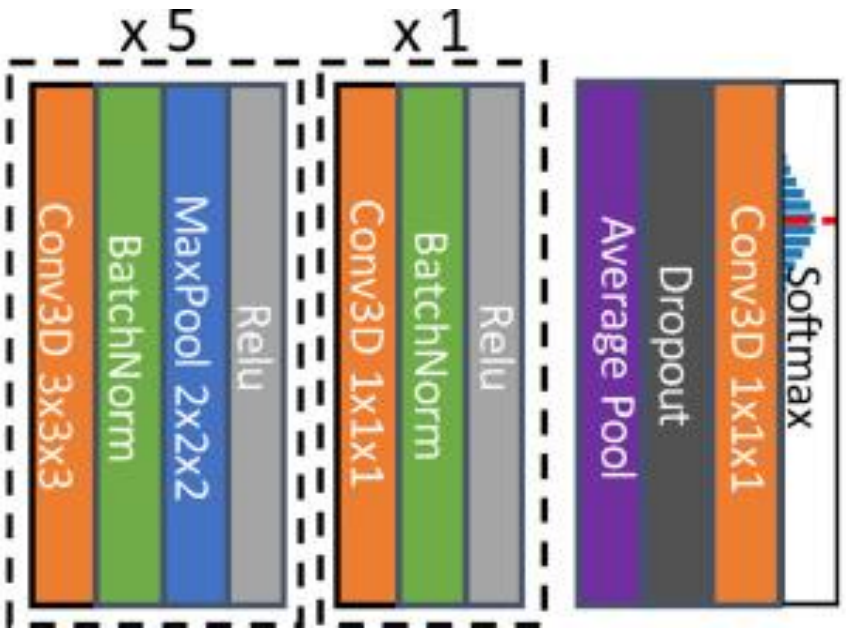
Open science: worth pursuing to boost the field ❤️

GitHub

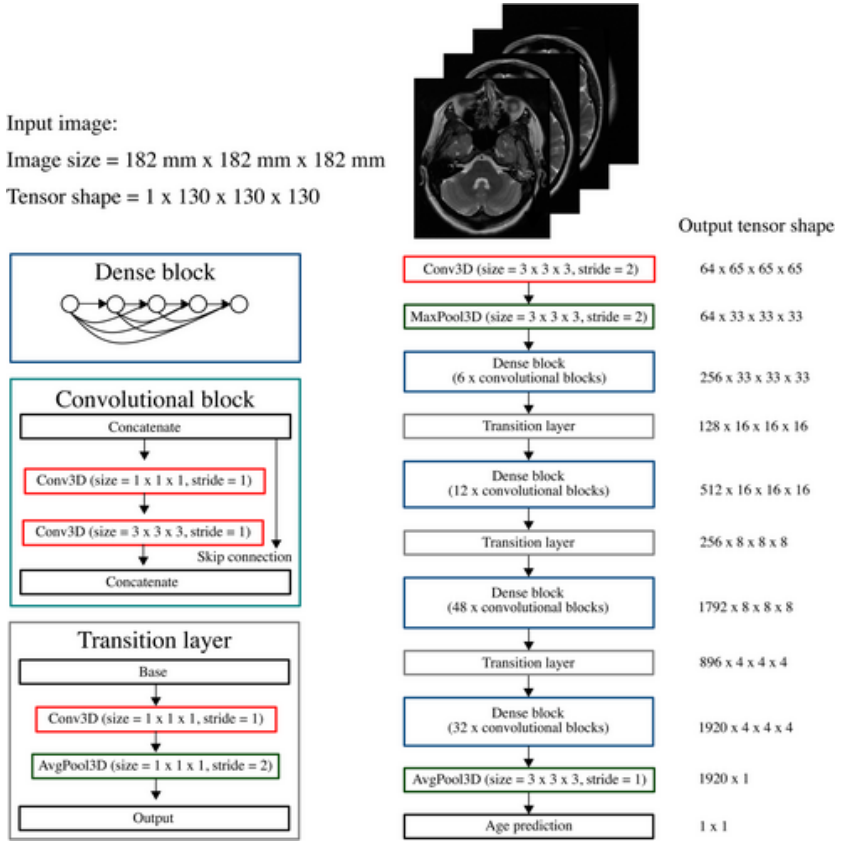
Hugging Face

Level 4: Deep CNNs

MAE_{HC}: 2.14 years



Peng et al. 2021. Medical Image Analysis



Wood et al. 2024. HBM.

MAE_{T1}: 3.38 years

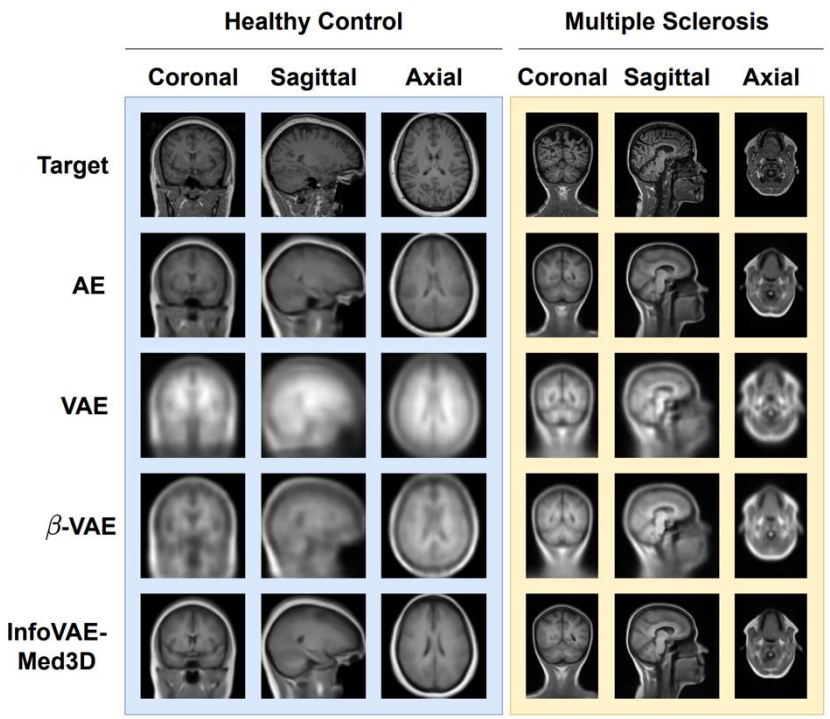
MAE_{T2}: 2.85 years

⚠

Image modality!

Level 5: Auto-encoders

- Phase 1: Self-supervised brain image generation
- Phase 2: Down-stream age-prediction task from latent space



Models	BrainAge		
	brain age		
	MAE ↓	R^2 ↑	RMSE ↓
AE	8.348	0.751	10.70
VAE	11.5675	0.425	16.249
β-VAE	9.957	0.648	12.820
InfoVAE-Med3D	**7.664**	**0.778**	**10.09**

Trinh Ngoc Huynh et al. 2025

Level 6: Foundational model hype?

Article | Open access | Published: 05 February 2026

A generalizable foundation model for analysis of human brain MRI

Divyanshu Tak, Biniam A. Garomsa, Anna Zapaishchykova, Tafadzwa L. Chaunzwa, Juan Carlos Climent Pardo, Zezhong Ye, John Zielke, Yashwanth Ravipati, Suraj Pai, Sri Vajapeyam, Maryam Mahootiha, Mitchell Parker, Luke R. G. Pike, Ceilidh Smith, Ariana M. Familiar, Kevin X. Liu, Sanjay Prabhu, Omar Arnaout, Pratiti Bandopadhayay, Ali Nabavizadeh, Sabine Mueller, Hugo JWL Aerts, Raymond Y. Huang, Tina Y. Poussaint & Benjamin H. Kann

Nature Neuroscience **29**, 945–956 (2026) | Cite this article

44k Accesses | **5** Citations | **364** Altmetric | Metrics

HUMAN BRAIN MAPPING
Open Access

RESEARCH ARTICLE | Open Access

Optimising brain age estimation through transfer learning: A suite of pre-trained foundation models for improved performance and generalisability in a clinical setting

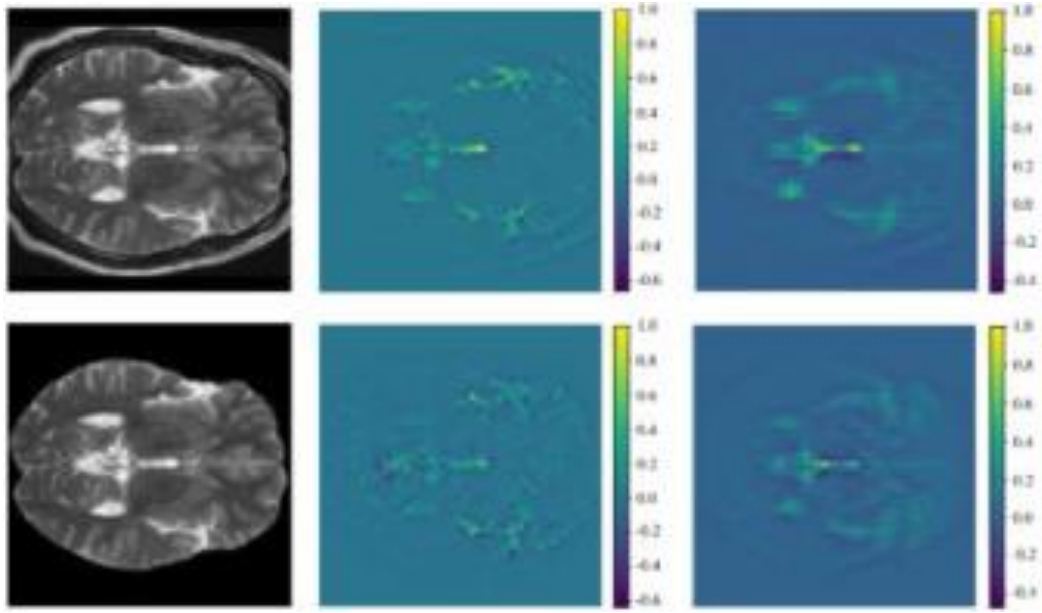
David A. Wood, Matthew Townend, Emily Guilhem, Sina Kafiabadi, Ahmed Hammam, Yiran Wei, Ayisha Al Busaidi, Asif Mazumder, Peter Sasieni, Gareth J. Barker, Sebastien Ourselin ... See all authors

First published: 04 March 2024 | https://doi.org/10.1002/hbm.26625 | VIEW METRICS

Tak et al. 2026, Nature Neuroscience || Wood et al. 2024. HBM.

Level 7: Back to simplicity?

- By ”asymptoting” the MAE=0, do we still know what aspects of brain aging we capture?



MRI Saliency Overlay

XAI technique used:
Guided backpropagation (image left), verified with occlusion sensitivity

↓

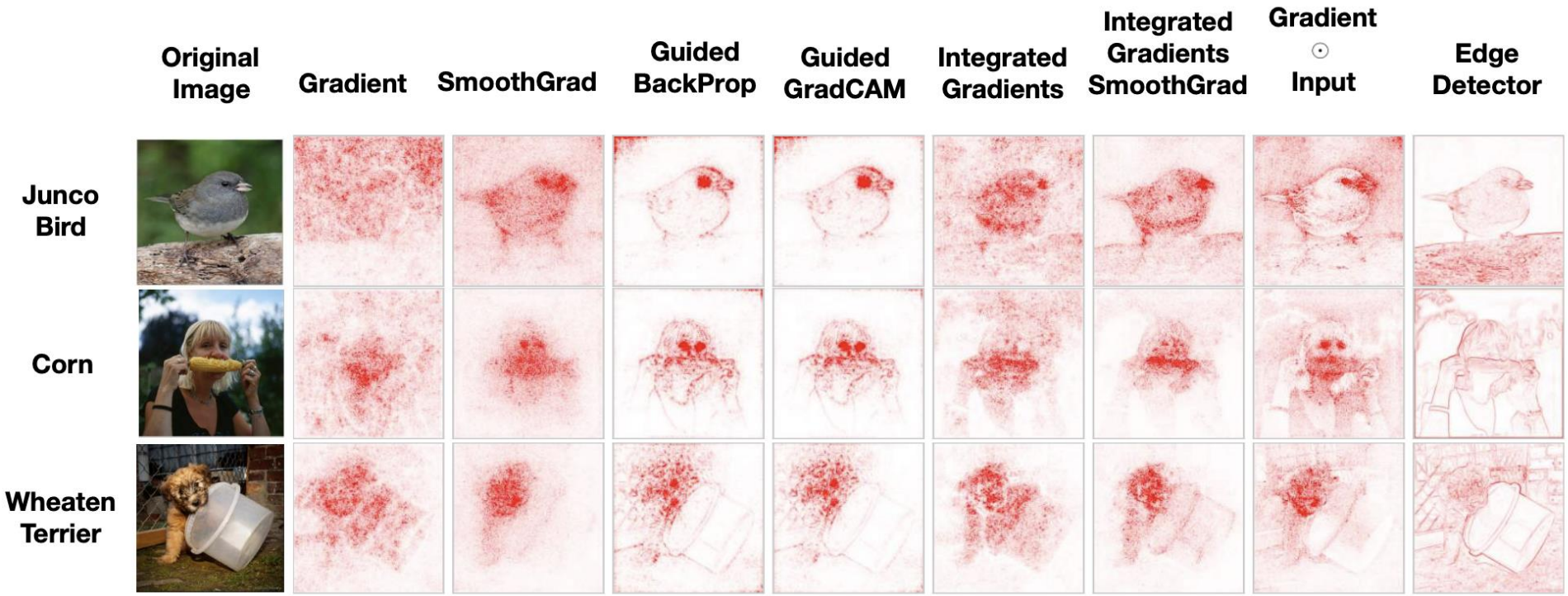
“saliency maps … primarily appear to be related to the cerebrospinal fluid spaces, such as the lateral ventricles”

⇣

Trust?

Wood et al. 2022. NeuroImage.

Explainability: Complicated edge detectors?



"We see that an edge detector produces outputs that are strikingly similar to the outputs of some saliency methods." and "Of the methods tested, Gradients & GradCAM pass the sanity checks, while Guided BackProp & Guided GradCAM are invariant to higher layer parameters; hence, fail."

Adebayo et al. 2018. 32nd NeurIPS 2018.

Which technical limitations do you see?

Mentimeter

Considerations

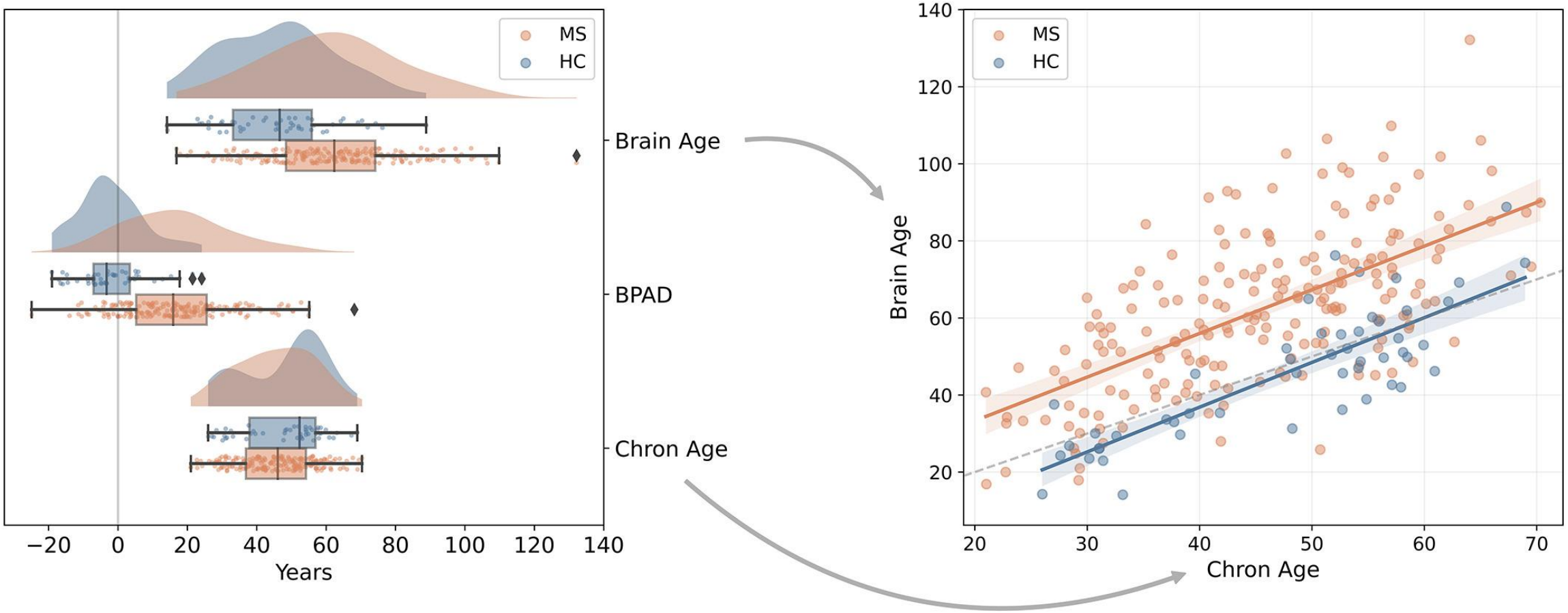
- What about other modalities?
  - E.g. T2w vs T1w in Wood et al. 2024
  - Functional imaging (EEG, MEG, FNIRS, …)
- Do we want to go to MAE = 0?
  - I invite you to ask Prof. Nagels. He will love that discussion ☺
- What aspect of aging do we want to capture?
  - Is explainability the bridge to the **clinical tale**?

Break

Part 2

The clinical tale

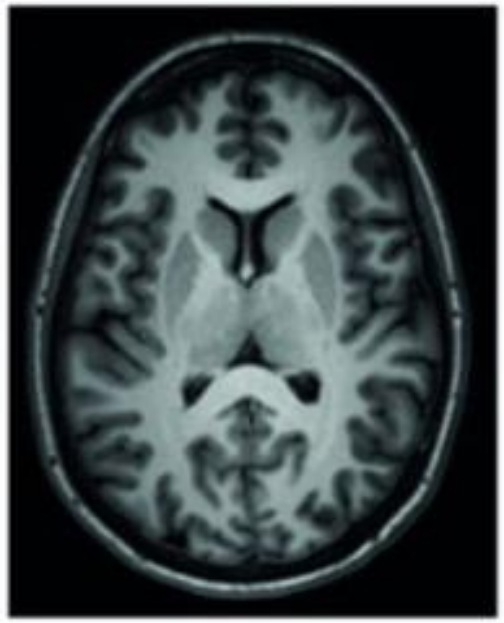
Brain age is overestimated in many disorders



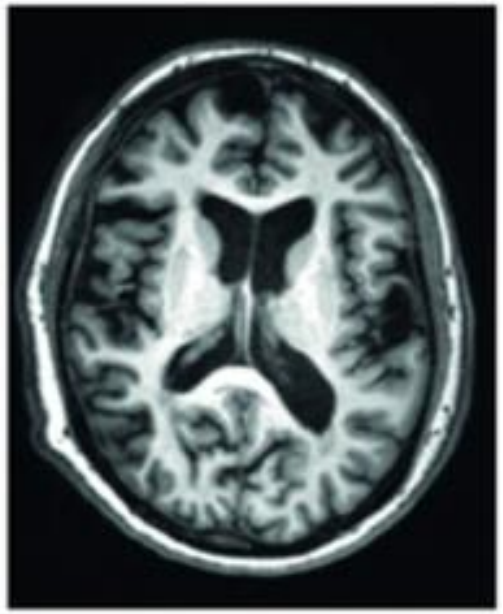
Denissen et al. 2022, Eur. J. Neurol.

Brain age is overestimated in many disorders

Healthy Control



PPMS



30.4	Age	49.0
29.6	Brain Age	63.9
-0.8	BPAD	+14.9

Image adapted from: Cole, James H., et al. Annals of neurology 88.1 (2020): 93-105.

Brain age is overestimated in many disorders

Higher BAG compared to controls in:

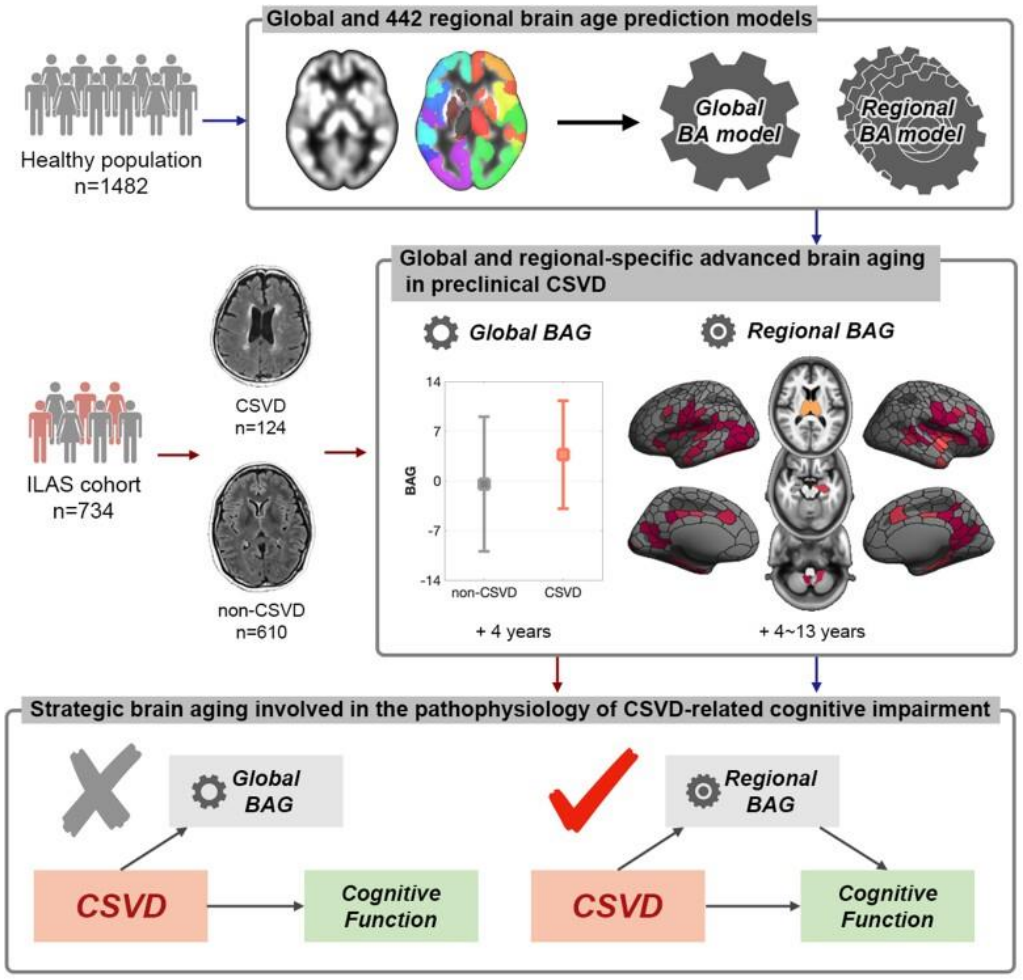
- Post-Traumatic Stress Disorder[1]
- Alzheimer's disease and MCI[2]
- Schizophrenia, Bipolar Disorder, Major Depression, Anxiety disorder[3]
- …

⚠ But how specific are they? Higher brain age = "abnormality"?

[1] Clausen et al. 2021. Brain Behav. || [2] Wittens et al. 2024. Alzheimers Res Ther.
[3] Seitz-Holland et al. 2024. Neurosci Biobehav Rev.

A case for “regional brain age”?

”Global brain-age gap failed to mediate the relationship between cerebral small vessel disease and any of the cognitive domains. In 89 regions with increased brain-age gap in the cerebral small vessel disease group, seven regional brain-age gaps were able to show significant mediation effects in cerebral small vessel disease-related cognitive impairment”



ILAS - I-Lan Longitudinal Aging Study; CSVD - ceerebral small vessel disease; BA - brain age; BAG - brain age gap

Lee et al. 2022. Brain Commun.

A case for “regional brain age”?

- “Experimental validation using UK Biobank data demonstrated that our framework successfully identified the thalamus (THA) and hippocampus (HIP) as core contributors to brain age prediction model decisions, highlighting their close associations with physiological aging.”[1]
- “These results suggest that regional brain age patterns reflect fundamental principles of cortical organization and behavior.”[2]
- ”Children with ADHD showed negative BAG across multiple regions (Cohen’s d: -0.12 to -0.08), while subcortical BAG emerged as a transdiagnostic indicator of younger-appearing brains”[3]

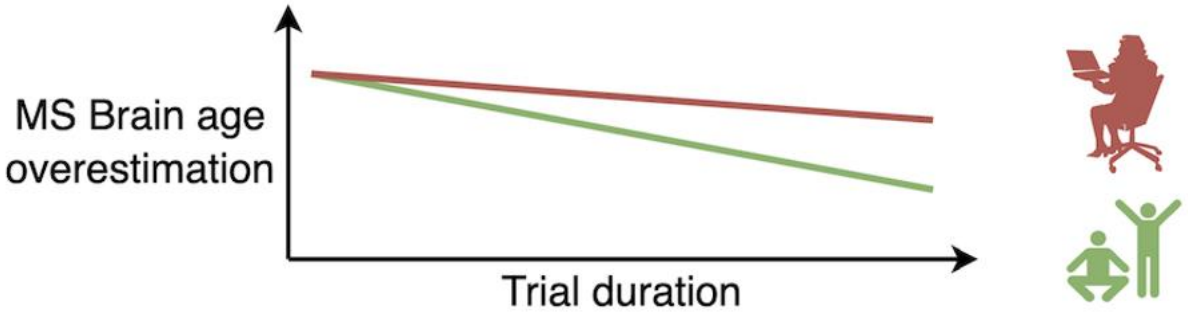
[1] Wu et al. 2025. Bioengineering || [2] Riccardi et al. 2025. Communications Biology.
[3] Gagnon et al. 2026

Can exercise reduce brain aging?

- Please reach out if you want to figure this out with me 🙂

Study	Population	Exercise type	Modality used	Conclusion
Gonzalez-Gomez 2026	Healthy	Resistance	fMRI	Exercise ↓ BAG
Yilmaz 2025	Schizophrenia	Balance	T1 MRI	No difference
Wan 2025	Healthy	Aerobic	T1 MRI	Exercise ↓ BAG
Wing 2024	Healthy	Combined	T1 MRI	No difference

My hypothesis for MS:



[1] Wu et al. 2025. Bioengineering || [2] Riccardi et al. 2025. Communications Biology.
[3] Gagnon et al. 2026

Considerations

- Brain age is not a reality in clinical practice
- Increasing evidence on factors that increase brain aging[1,2]
- Unclear how to counteract accelerated brain aging
  - Especially relevant in an aging population with more dementia

[1] Legaz et al. 2026. Nature Medicine. || [2] Wittens et al. 2024. Alzheimers Res Ther.

Would you convince doctors to use brain age?

Mentimeter

Image sources

- https://www.someecards.com/usercards/viewcard/i-take-comfort-in-knowing-that-people-younger-than-me-look-older-than-me-2826e/
- https://www.pinterest.com/pin/202802789457523611/

References

Adebayo, Julius, et al. "Sanity checks for saliency maps." *Advances in neural information processing systems* 31 (2018).

Clausen, Ashley N., et al. "Assessment of brain age in posttraumatic stress disorder: Findings from the ENIGMA PTSD and brain age working groups." *Brain and behavior* 12.1 (2022): e2413.

Cole, James H., et al. "Longitudinal assessment of multiple sclerosis with the brain-age paradigm." *Annals of neurology* 88.1 (2020): 93-105.

Denissen, Stijn, et al. "Brain age as a surrogate marker for cognitive performance in multiple sclerosis." *European journal of neurology* 29.10 (2022): 3039-3049.

Denissen, Stijn, et al. "Real-world federated learning for brain imaging scientists." *Frontiers in digital health* 8 (2026): 1691088.

Gagnon, Anthony, et al. "Regional brain age deviations reveal divergent developmental pathways in youth." *bioRxiv* (2025): 2025-11.

Huynh, Trinh Ngoc, et al. "Latent Representation Learning from 3D Brain MRI for Interpretable Prediction in Multiple Sclerosis." *arXiv preprint arXiv:2510.00051* (2025).

Lee, Pei-Lin, et al. "Regional rather than global brain age mediates cognitive function in cerebral small vessel disease." *Brain Communications* 4.5 (2022): fcac233.

Legaz, Agustina, et al. "The exposome of brain aging across 34 countries." *Nature medicine* (2026): 1-14.

Peng, Han, et al. "Accurate brain age prediction with lightweight deep neural networks." *Medical image analysis* 68 (2021): 101871.

Riccardi, Nicholas, et al. "Distinct brain age gradients across the adult lifespan reflect diverse neurobiological hierarchies." *Communications Biology* 8.1 (2025): 802.

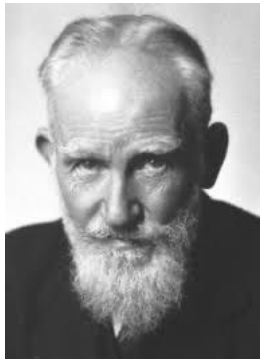
Seitz-Holland, Johanna, et al. "BrainAGE, brain health, and mental disorders: A systematic review." *Neuroscience & Biobehavioral Reviews* 159 (2024): 105581.

Tak, Divyanshu, et al. "A generalizable foundation model for analysis of human brain MRI." *Nature Neuroscience* (2026): 1-12.

Wittens, Mandy MJ, et al. "Brain age as a biomarker for pathological versus healthy ageing–a REMEMBER study." *Alzheimer's research & therapy* 16.1 (2024): 128.

Wood, David A., et al. "Accurate brain-age models for routine clinical MRI examinations." *Neuroimage* 249 (2022): 118871.

Wood, David A., et al. "Optimising brain age estimation through transfer learning: A suite of pre-trained foundation models for improved performance and generalisability in a clinical setting." *Human Brain Mapping* 45.4 (2024): e26625.

Wu, Yutong, et al. "Regional Brain Aging Disparity Index: Region-Specific Brain Aging State Index for Neurodegenerative Diseases and Chronic Disease Specificity." *Bioengineering* 12.6 (2025): 607.

We don’t stop playing because we grow old; we grow old because we stop playing” – George Bernard Shaw